CITY AND COUNTY OF SAN FRANCISCO
MUNICIPAL PORTFOLIO

SF Rec Park Permits & Reservations Analytics Engine

Operational Pipeline, Throughput Analysis, and Interactive Dashboard
for the 1820 Junior Administrative Analyst Portfolio

Prepared by: Sileshi Hirpa

Target department: San Francisco Recreation and Park Department

Office: Permits & Reservations

Report date: 2026-10-23

Data snapshot: 2026-09-30

Independent portfolio study. All permit transactions and operational results are synthetic.
Not an official City report or representation of departmental performance.

Table of Contents

Abstract	Page 2
Motivation and urban context	Page 3
Problem statement	Page 3
Methodology and architecture	Page 4
Results: synthetic operational performance	Page 5
SLA framework and metric definitions	Page 6
Limitations	Page 7
Conclusions	Page 7
Recommendations	Page 7
Appendix: facility reference	Page 8
Sources, deployment and reproducibility	Page 9
Visual appendix: executive summary	Page 10
Visual appendix: throughput and bottlenecks	Page 11
Visual appendix: permit explorer	Page 12
Visual appendix: facility demand map	Page 13

Abstract

This portfolio demonstrates a reproducible Python, Pandas, SQLite, and Streamlit workflow for municipal permit operations. It models 12,000 fictional applications as of 2026-09-30, separates completed-case turnaround from pending-case age, and exposes review delays through a filterable dashboard. Authentic DataSF property coordinates provide geographic context, while facility assignments remain simulated.

The snapshot yields 10.09 calendar days of average resolved turnaround, 159 pending applications, and 17 overdue pending cases under assumed category targets. These outputs demonstrate analytical methods, not measured SF Rec Park operations.

Core objective: Make administrative queue measures explicit, reproducible, and understandable so an analyst can distinguish active backlog from historical late decisions.

Project Motivation & Urban Context

The project is motivated by public service engagement and everyday observations of how residents and visitors use San Francisco. Rideshare conversations about neighborhoods, destinations, and access to parks provide a practical starting point for asking how public assets and administrative services support urban life. These observations are motivation, not a representative survey or evidence of permit demand.

For an 1820 Junior Administrative Analyst portfolio, the emphasis is on defensible calculations, data quality, transparent assumptions, and clear operational reporting. The application connects transaction-level records to summaries that can support questions about service delivery, queue monitoring, and workload distribution.

Problem Statement

An analyst needs to distinguish volume from backlog, pending age from completed turnaround, and collected revenue from quoted fees. Mixing those measures can hide active queues or misstate performance. This simulation builds a consistent snapshot that can be validated, queried, filtered, and explained without exposing applicant information.

12,000	10.09 days	4 zones
Synthetic applications	Average resolved turnaround	Invented analytical groupings

Operational questions: Which categories have longer completed review cycles? Which pending applications exceed their assumed targets? How are modeled fees distributed? How do filters change the interpretation of executive measures?

Interpretation boundary: The tool demonstrates a decision-support workflow. It does not establish actual department delays, staffing requirements, booking eligibility, customer outcomes, or facility utilization.

Methodology & Data Architecture

Stage	Implementation	Output
Generate	data_generator.py	Fictional CSV + seed/version manifest
Validate and enrich	etl.py	Cleaned snapshot, dates, durations, flags
Store and aggregate	SQLite	Permits table, indexes, KPI views
Reference and map	appendix_data.py	Sourced property points + simulated assignments
Explore and report	app.py / generate_report.py	Filtered dashboard and this PDF

The generator uses a fixed seed and observation cutoff. Seasonal intake weights, category mix, review-time distributions, fees, approval rates, and analytical zones are declared assumptions. Special events have a deliberately longer review distribution, and the South zone has a modeled lag multiplier. Later comparisons therefore recover design choices rather than independent empirical findings.

The ETL trims text, validates types and required fields, enforces unique synthetic IDs and a single snapshot, and rejects impossible chronology. Approval and rejection dates must agree with status; pending records have no decision date. Revenue follows the explicit full-payment-on-approval assumption. A temporary SQLite build replaces the prior database only after a successful load.

All durations are calendar days. Completed turnaround is decision date minus application date. Pending age is snapshot date minus application date. A bottleneck is a pending case whose age is strictly greater than its category target. Decisions exactly on target are compliant; historical late decisions remain a separate measure.

SQLite views expose executive, category, zone, category-zone, and decision-month summaries. Dashboard filters define application cohorts. The facility layer selects one authentic property within each invented zone using a stable hash of permit ID; it changes neither transaction counts nor revenue.

Results & Synthetic Operational Performance

Indicator	Computed result
Total / resolved applications	12,000 / 11,841
Approved / rejected	10,355 / 1,486
Average resolved turnaround	10.09 calendar days
Modeled revenue collected	$2,943,190.26
Pending / backlog percentage	159 / 1.325%
Overdue pending / share of pending	17 / 10.69%
Historical late decisions	1,691
Resolved within assumed target	85.72%

Category comparison: resolved cases only. Counts and revenue include all statuses in each category.

Category	Permits	Avg days	Overdue pending	Revenue (USD)
Special Event	1,211	21.30	4	$1,353,448.10
Clubhouse	2,104	12.72	2	$614,493.10
Athletic Field	4,749	9.74	9	$685,005.74
Picnic Area	3,936	5.74	2	$290,243.32

Zone comparison: invented analytical groupings, not official department districts.

Zone	Permits	Avg resolved days	Overdue pending
East	3,556	9.51	2
North	3,150	9.45	3
South	2,954	11.94	10
West	2,340	9.51	2

SLA Framework & Metric Definitions

Current baseline: service-category targets measured in calendar days. The database has date-only records and does not model customer profiles, intake acknowledgments, priority escalation, business hours, holidays, or portal availability.

Permit category	Assumed calendar-day target
Athletic Field	14
Clubhouse	21
Picnic Area	10
Special Event	30

Backlog percentage: pending / total applications x 100.
Overdue pending percentage: overdue pending / pending x 100.
Within-target compliance: resolved on or before target / resolved x 100.
Revenue: sum of collected integer US cents, divided by 100 for dollars.
Undefined measures: empty denominators are NULL / N/A, not zero.

Proposed Customer-Based Framework

The proposed framework would layer customer-profile differentiation onto a baseline service model. The following user-proposed priorities are future design objectives, not verified SF Rec Park commitments and not implemented as current business-day targets.

Priority	Initial response	Resolution
P1: special events / urgent	4 business hours	5 business days
P2: athletic fields	24 business hours	10 business days
P3: picnic / general	48 business hours	15 business days

Additional proposed objectives are a 3% backlog ceiling, a 14-day stale threshold, and 99.9% portal uptime. Implementing them requires agreed customer/priority mappings, clubhouse treatment, a holiday and business-hours calendar, response event timestamps, and independent uptime monitoring. A priority label alone is not customer-based governance.

Limitations

1. Synthetic parameters are not calibrated to actual operations. No inference about real staffing, fees, bottlenecks, customer groups, or service quality is supported.
2. Pending cases are right-censored at the snapshot. Resolved-only turnaround can understate eventual processing time for recent cohorts.
3. Revenue assumes full collection on approval and omits refunds, waivers, partial payment, and accounting adjustments.
4. Facility assignment is illustrative and does not validate permit eligibility, available amenities, capacity, occupancy, or actual demand. The 19 reference sites are a curated subset.
5. The project does not establish legal restrictions on all transaction records. Availability and disclosure rules require field-specific source verification.
6. No business-day response compliance, uptime, causal impact, or forecast accuracy is measured.

Conclusions

The completed workflow reconciles 12,000 synthetic transactions across CSV, SQLite, dashboard summaries, and facility demand totals. It distinguishes 17 actionable overdue pending records from 1,691 historical late decisions. Its portfolio value is a reproducible, auditable method for operational analysis, not evidence of an actual departmental performance gap.

Recommendations

Standardize intake: Pilot document checklists and consistent acknowledgment events. Measure completeness and follow-up rates before proposing a quantified benefit.
Review the active queue: Use overdue age and case context to support staff triage. Do not infer urgency or regulatory complexity from elapsed days alone.
Define governance: Agree on service targets, customer tiers, business calendars, exception rules, metric owners, and a reporting cadence before operational adoption.
Evaluate capacity carefully: Gather actual reservation slots, closures, amenities, and occupancy before recommending dynamic facility or staffing allocation.
Validate with a baseline: Compare a small pilot against documented outcomes. No 20% reduction or other causal gain is claimed by this simulation.

Appendix: Authentic Facility Reference

Curated DataSF properties. Names, IDs and coordinates are source fields. Analytical zones are invented; points represent properties, not reservable entrances. Golden Gate Park retains its seven source sections.

Property ID / source name	Latitude	Longitude	Modeled zone
112: Yacht Harbor and Marina Green	37.807963	-122.441198	North
125: Hamilton Rec Center	37.784424	-122.436016	North
132: Moscone Rec Center	37.802131	-122.433122	North
150: Joe DiMaggio Playground	37.802497	-122.412169	East
164: Crocker Amazon Playground	37.714042	-122.429288	South
172: Richmond Rec Center	37.783275	-122.477990	West
28: John McLaren Park	37.718097	-122.419080	South
31: St. Mary's Rec Center	37.733675	-122.420671	South
52: Potrero Hill Rec Center	37.756141	-122.396940	East
63: Mission Dolores Park	37.759755	-122.427128	East
66: Balboa Park	37.724960	-122.444625	South
756665: Golden Gate Park - Section 1	37.771391	-122.455294	West
756666: Golden Gate Park - Section 2	37.770904	-122.466033	West
756667: Golden Gate Park - Section 3	37.767268	-122.470391	West
756668: Golden Gate Park - Section 4	37.768894	-122.479092	West
756669: Golden Gate Park - Section 5	37.769409	-122.491303	West
756670: Golden Gate Park - Section 6	37.767730	-122.504238	West
93: Sunset Rec Center	37.756594	-122.486857	West
957231: Golden Gate Park - Section 7	37.768455	-122.460759	West

Source: DataSF Recreation and Parks Properties (gtr9-ntp6). Retrieved 2026-10-23. Source precision is preserved in the bundled CSV; coordinates above are rounded to six decimals.

Explore the deployed facility map in the live dashboard; source records and allocation code are in the public GitHub repository. The visual appendix begins on page 10.

Sources, Deployment & Reproducibility

Live dashboard: https://sf-rec-park-analytics.streamlit.app/
Public source repository: https://github.com/sileshith/sf-rec-park-analytics
Public access and deployed views verified on 2026-10-23.

Public reference data: DataSF Recreation and Parks Properties. The inventory describes land owned and/or maintained by the department. A 19-record excerpt is preserved with the retrieval date and full-download hash. The catalog lists PDDL 1.0 for the source data; repository code is MIT licensed.

Data.gov catalog and source metadata
Public Domain Dedication and License 1.0
Streamlit documentation
ReportLab documentation

Transaction provenance: data/permits.manifest.json records generator parameters and NumPy/Pandas versions. For exact synthetic regeneration, use those recorded versions. Application dependency ranges permit updates; the committed CSV and database are the stable portfolio snapshot.

Database provenance: schema version 1; snapshot 2026-09-30; source CSV SHA-256:

```
cc31fbeed28e571701dbb4a3019ddb81c3a78cae052fbf79999ee04964afcae0
```

Rebuild from repository root:
1. Install requirements.txt and requirements-report.txt.
2. Run python etl.py to validate and rebuild SQLite from the committed CSV.
3. Run python -m unittest discover -s tests -v.
4. Run python generate_report.py.
5. Run streamlit run app.py to inspect the same snapshot.

Dashboard coverage: Executive summary; throughput and bottlenecks; filterable record explorer; illustrative facility demand map; SLA framework and definitions. Filters use application dates and apply to all analysis tabs. Map tiles require internet; SQLite and reference tables are local.

Deployment status: The dashboard is publicly deployed on Streamlit Community Cloud from main / app.py using Python 3.11. The bundled SQLite snapshot loads without a cloud ETL job. GitHub Actions has passed on Python 3.11 and 3.12, including the report build. Docker configurations remain available but container execution has not been verified. Inactive cloud apps may require a short wake-up period.

Visual Appendix 1: Executive Summary

Captured from the public deployment on 2026-10-23. The sidebar includes the final PDF download button. All permit records are synthetic.

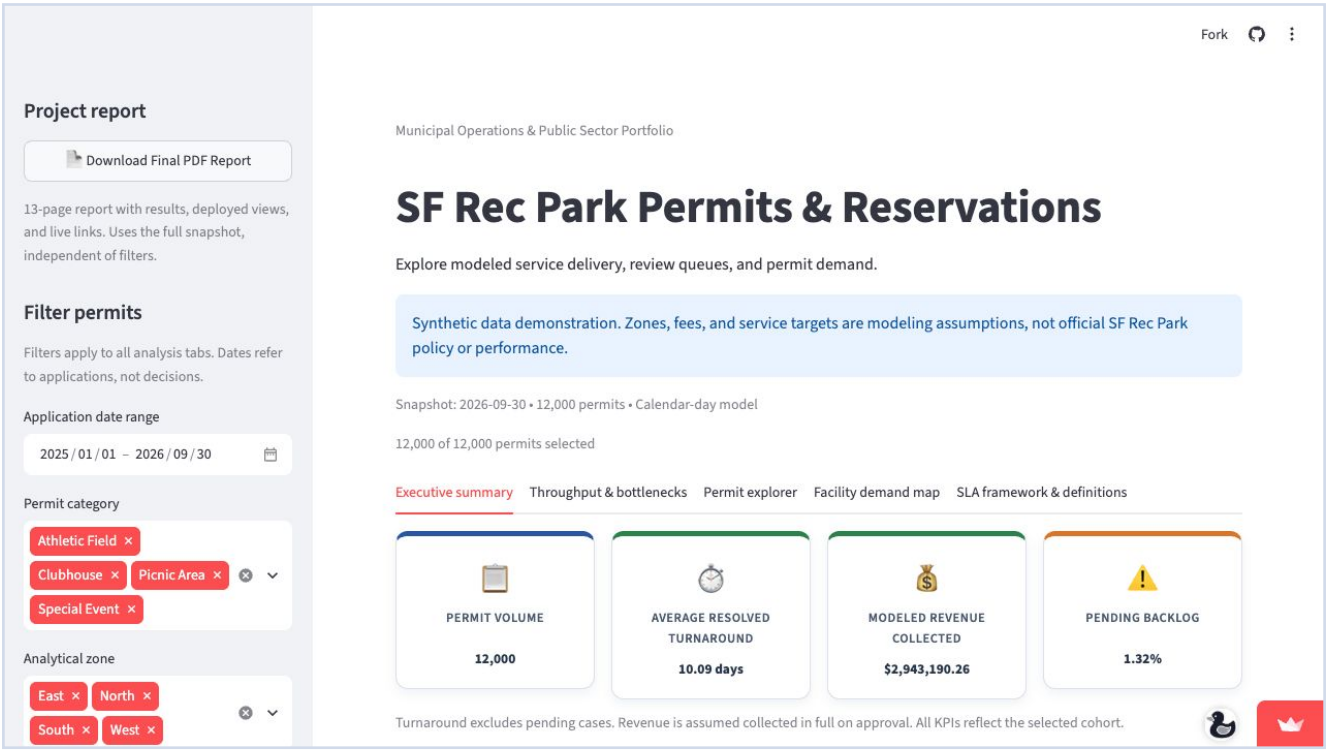


Figure 1. The executive view summarizes the full synthetic snapshot, with all category, zone, and status filters selected.

How to read this view

Read resolved turnaround separately from pending age. The revenue figure is modeled collection on approval, not an observed department receipt. Backlog is rounded to two decimals on screen; the precise full-snapshot value is 1.325%.

Live view: https://sf-rec-park-analytics.streamlit.app/
Source and reproducibility: GitHub repository

Original image: assets/screenshots/deployed/executive-summary.png. Capture provenance and SHA-256 values are recorded in capture_manifest.json in that directory. Click the screenshot to open the live application.

Visual Appendix 2: Throughput & Bottlenecks

Captured from the public deployment on 2026-10-23. The sidebar includes the final PDF download button. All permit records are synthetic.



Figure 2. The category comparison shows completed turnaround alongside pending age. All four synthetic permit categories are included.

How to read this view

Longer special-event reviews are a generator assumption. This comparison is descriptive of modeled records. The active queue below the chart flags pending records strictly past their assumed category targets.

Live view: https://sf-rec-park-analytics.streamlit.app/
Source and reproducibility: GitHub repository

Original image: assets/screenshots/deployed/throughput-bottlenecks.png. Capture provenance and SHA-256 values are recorded in capture_manifest.json in that directory. Click the screenshot to open the live application.

Visual Appendix 3: Permit Explorer

Captured from the public deployment on 2026-10-23. The sidebar includes the final PDF download button. All permit records are synthetic.

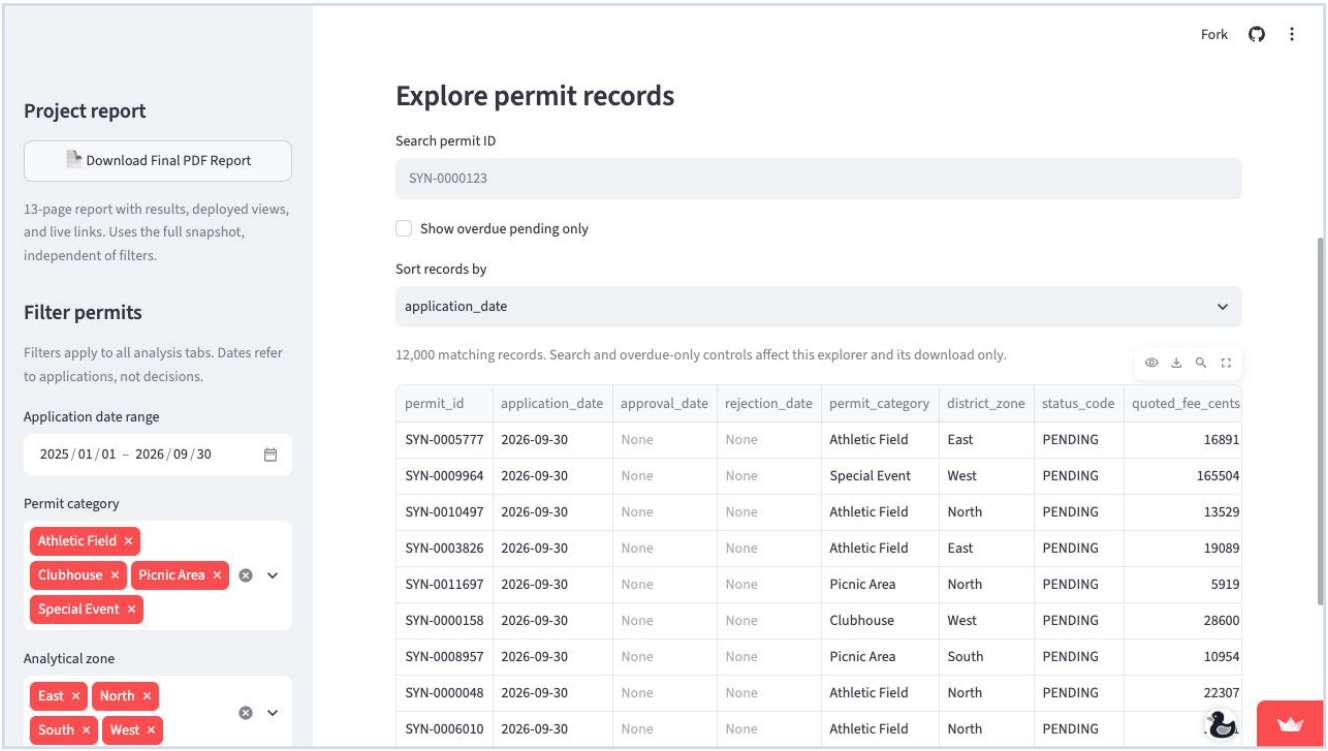


Figure 3. The record explorer shows the full 12,000-record selection with an empty ID search and the overdue-only toggle off.

How to read this view

Application dates define the selected cohort. Recent records at the snapshot can remain pending, with blank decision fields. Search and sorting affect the explorer and its CSV export, while the sidebar filters affect all analysis tabs.

Live view: https://sf-rec-park-analytics.streamlit.app/
Source and reproducibility: GitHub repository

Original image: assets/screenshots/deployed/permit-explorer.png. Capture provenance and SHA-256 values are recorded in capture_manifest.json in that directory. Click the screenshot to open the live application.

Visual Appendix 4: Facility Demand Map

Captured from the public deployment on 2026-10-23. The sidebar includes the final PDF download button. All permit records are synthetic.

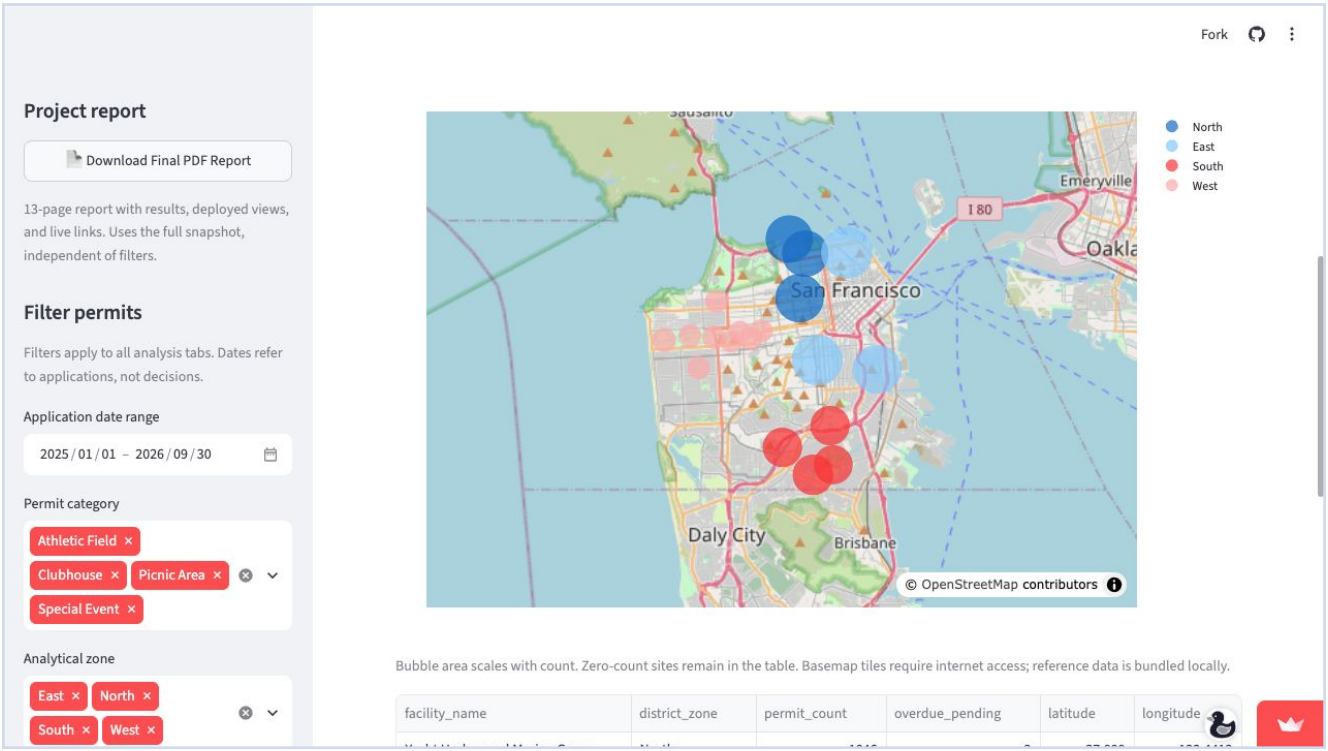


Figure 4. The map displays assigned synthetic permit counts at authentic DataSF property coordinates. All sidebar filters are selected.

How to read this view

Bubble area represents assigned count. Colors identify invented analytical zones, not official districts. The stable allocation is not evidence of real bookings, category eligibility, capacity, or utilization. Basemap attribution: OpenStreetMap contributors.

Live view: https://sf-rec-park-analytics.streamlit.app/
Source and reproducibility: GitHub repository

Original image: assets/screenshots/deployed/facility-demand.png. Capture provenance and SHA-256 values are recorded in capture_manifest.json in that directory. Click the screenshot to open the live application.